

A fermion insertion on the Ramond tube of the theta block

Separate working notes

September 10, 2026

We derive a fermion-inserted double-*Virasoro* representation of the NS–R–R theta block with $\eta\eta' = -1$, using the states, BPZ pairing, branching coefficients and spin signs of `Human Notes/SCblock.tex`. Both *Virasoro* factors in the enlarged block are evaluated by central-charge recursion. The auxiliary factor is evaluated directly by fermion-state sewing; its Ising decomposition identifies the irreducible representations being propagated. One contour phase missing from the current definition of $\hat{\rho}$ is made explicit below. It is necessary for the branching and physical sewing conventions to give the stated convolution.

1 The additional sphere and the operator on the tube

The two original spheres have Neveu–Schwarz punctures at infinity and Ramond punctures at one and zero. We replace the second edge by a sphere with coordinate z_3 , with sewing equations

$$(z_1 - 1)z_3 = \hat{q}_{2,1}, \quad (z_2 - 1)z_3^{-1} = \hat{q}_{2,2}. \quad (1.1)$$

Eliminating z_3 gives

$$(z_1 - 1)(z_2 - 1) = q_2, \quad q_2 = \hat{q}_{2,1}\hat{q}_{2,2}. \quad (1.2)$$

Thus this insertion introduces a marked point on the second tube. Moving it changes the two sewing parameters while preserving their product. We choose the transported Ramond spin frame on the additional sphere so that the lifts satisfy $\eta_2 = \hat{\eta}_{2,1}\hat{\eta}_{2,2}$. This fixes the square-root transport convention in addition to the coordinate relation (1.2).

In the physical irreducible NS vacuum module, $G_{-1/2}\mathbf{1} = 0$. The normalization in the human notes consequently gives

$$v_{1/2}(P = Q/2) = Q\psi_{-1/2}\mathbf{1}, \quad V[v_{1/2}(Q/2)](1) = Q\psi(1), \quad Q = b + b^{-1}. \quad (1.3)$$

The field has the physical identity as its SCA factor. The insertion we use is the radially ordered tube operator

$$\Theta V[v_{1/2}(Q/2)](1) = Q\Theta\psi(1). \quad (1.4)$$

Here Θ acts on the Ramond module on the outgoing side of the field; it is not a prescription to apply a Ramond zero mode to an NS state.

For a chain Ψ_{-B} of nonzero Ramond fermion modes, define the odd Ramond intertwiner by

$$\Theta(\Psi_{-B}u^b) = (-1)^{\#B+b+1}\Psi_{-B}u^{1-b}. \quad (1.5)$$

The auxiliary conventions are $\psi_0 u^b = u^{1-b}/\sqrt{2}$, $\langle u^b | u^b \rangle = (-1)^b$ and $\psi_r^\dagger = -\psi_{-r}$. Directly from the displayed definition,

$$\Theta^2 = -1, \quad \{\Theta, \psi_r\} = 0, \quad \{\Theta, G_r\} = 0, \quad [\Theta, L_m] = [\Theta, L_m^{(1)}] = [\Theta, L_m^{(2)}] = 0. \quad (1.6)$$

The anticommutator with G_r uses the graded tensor product. The product $\Theta\psi$ is even and acts only on the auxiliary factor. Its zero-mode part acts diagonally:

$$Q\Theta\psi_0(\Psi_{-B}u^b) = \frac{Q}{\sqrt{2}}(-1)^b\Psi_{-B}u^b. \quad (1.7)$$

The signs from moving ψ_0 past the nonzero modes and from Θ cancel. This ground-state sign makes recovery of the $\eta\eta' = -1$ sector possible; the sector identity and inverse are derived below. The uninserted auxiliary block annihilates that sector under convolution. The full field also contains nonzero modes: we must not replace $\Theta\psi(1)$ by $\Theta\psi_0$ inside a split-tube block.

2 The double-Virasoro decomposition

The physical parameters and central charge are

$$h_1 = \frac{Q^2/4 - P_1^2}{2}, \quad \beta_2 = \frac{P_2}{\sqrt{2}}, \quad \beta_3 = \frac{P_3}{\sqrt{2}}, \quad c = \frac{3}{2} + 3Q^2. \quad (2.1)$$

Here p_1 is the intrinsic parity of the NS primary and f is the relative outer three-point label used in the human notes. In particular, the allowed descendant parities satisfy the selection rule displayed in (2.8). We assume $b \neq 0$, $b^2 \neq 1$ for the embedding and $Q \neq 0$ for the chosen insertion and its sector inverse. At isolated momentum poles the complete expressions are understood by meromorphic continuation, rather than by separately evaluating singular summands.

The internal double-Virasoro weights are

$$\begin{aligned} h_{j,n}^{(1)} &= \frac{Q^2/4 - (P_j + 2nb)^2}{2(1 - b^2)}, & c^{(1)} &= 1 + \frac{3Q^2}{1 - b^2}, \\ h_{j,n}^{(2)} &= \frac{Q^2/4 - (P_j + 2n/b)^2}{2(1 - b^{-2})}, & c^{(2)} &= 1 + \frac{3Q^2}{1 - b^{-2}}. \end{aligned} \quad (2.2)$$

The NS labels lie in $\frac{1}{2}\mathbb{Z}$ and contribute level $2n^2$. The Ramond labels lie in $\frac{1}{2}\mathbb{Z} + \frac{1}{4}$ and contribute level $2n^2 - \frac{1}{8}$. The latter subtraction measures level relative to the Ramond ground states, including the auxiliary ground weight $1/16$. The two weights of the inserted field are

$$h_\psi^{(1)} = -\frac{1 + 2b^2}{2(1 - b^2)}, \quad h_\psi^{(2)} = \frac{b^2 + 2}{2(1 - b^2)}. \quad (2.3)$$

Their sum is $1/2$. Its two level-two degenerate fusion rules require

$$n' = n + \frac{1}{2} \quad \text{or} \quad n' = n - \frac{1}{2}. \quad (2.4)$$

The odd field reverses the Ramond parity and Θ reverses it back. The two parts of the split edge therefore carry the same parity α . The outer three-point forms retain the same label f .

2.1 The outer vertex convention

The physical forms $\rho_f^{(\eta)}$ use the Ramond BPZ convention of the human notes. The free-field branching forms require the following contour phase when they are expressed in that physical tensor basis:

$$\begin{aligned} & \hat{\rho}_f^{(\eta)}(\Psi_{-A}\mathbf{L}_{-A}\phi_1, \Psi_{-B}u^b \otimes \mathbb{L}_{-B}w_2^\alpha, \Psi_{-C}u^c \otimes \mathbb{L}_{-C}w_3^\gamma) \\ &= (-i)^{A \bmod 2} (-1)^{f(A+B+|\alpha|)+AA+(B+|\alpha|+p_1)(C+c)} \\ & \quad \times \rho_F(\Psi_{-A}\mathbf{1}, \Psi_{-B}u^b, \Psi_{-C}u^c) \rho_f^{(\eta)}(\mathbf{L}_{-A}\phi_1, \mathbb{L}_{-B}w_2^\alpha, \mathbb{L}_{-C}w_3^\gamma). \end{aligned} \quad (2.5)$$

In (2.5), $|\alpha|$ is the parity of the physical Ramond ground w_2^α ; in (2.8), $\alpha, \gamma \in \{0, 1\}$ label the total parities of the enlarged Ramond primaries. Descendant indices in sign exponents count fermionic generators; their absolute values denote levels. The factor $(-i)^{A \bmod 2} (-1)^{f(A+B+|\alpha|)}$ follows by comparing the NS–R–R supercurrent Ward identities in the free-field and physical frames, after the Ramond ground basis is transported to its physical BPZ normalization. The implementation’s three-point label must also be transported as $\eta_{\text{native}} = (-1)^f \eta$. For example, in the odd form the relative coefficients of β_2 and β_3 in $\rho(G_{-1/2}\phi, w^+, w^+)$ have the wrong relative sign if this label transport is omitted. The NS supercurrent step changes the contour factor by $-i(-1)^{A+f}$, while flipping the second Ramond parity changes it by $(-1)^f$; substitution of these ratios reproduces the three physical supercurrent Ward recursions. The transport also has a direct double-Virasoro interpretation: in this frame the physical supercurrent and auxiliary fermion contours at infinity have multipliers $-i$ and $+i$, so their product is $+1$, as required for the integer-spin field $:G\psi:$ entering $L^{(1,2)}$.

Equation (2.5) supplies a correction to the current human-notes factorization of $\hat{\rho}$, which omits this contour phase while retaining $(-1)^{A+A}$ in the enlarged sewing sum. Without the phase, an unwanted $(-1)^A$ remains in the physical factor. We use (2.5) throughout, including in the branching calculation.

2.2 Sewing the four internal edges

We first define the ordinary punctured Virasoro block, keeping the order of the three punctures $(\infty, 1, 0)$ explicit. In the following formula the four internal weights are (h_1, h_L, h_R, h_3) ; the additional weight h_ψ is external. This use of L, R only specifies which half of the second tube

is meant. With each inverse Gram matrix restricted to equal levels,

$$\begin{aligned}
& F(h_1, h_L, h_R, h_3; h_\psi; c \mid q_1, \hat{q}_{2,1}, \hat{q}_{2,2}, q_3) \\
&= \sum_{\substack{|A|=|A'|, |B|=|B'| \\ |C|=|C'|, |D|=|D'|}} q_1^{|A|} \hat{q}_{2,1}^{|B|} \hat{q}_{2,2}^{|C|} q_3^{|D|} G_{h_1}^{AA'} G_{h_L}^{BB'} G_{h_R}^{CC'} G_{h_3}^{DD'} \\
&\quad \times \rho(L_{-A}\nu_{h_1}, L_{-B}\nu_{h_L}, L_{-D}\nu_{h_3}) \rho(L_{-A'}\nu_{h_1}, L_{-C}\nu_{h_R}, L_{-D'}\nu_{h_3}) \\
&\quad \times \rho(L_{-C'}\nu_{h_R}, \nu_{h_\psi}, L_{-B'}\nu_{h_L}).
\end{aligned} \tag{2.6}$$

The leading coefficient is one. There is no constraint $|B| = |C|$: the insertion can change the auxiliary level. The displayed Gram contraction defines the block; the production computation evaluates it by c -recursion, not by constructing these Gram matrices.

The middle primary matrix element needed for sewing is

$$\langle v_{2,n'}^\alpha | \Theta V[v_{1/2}(Q/2)](1) | v_{2,n}^\alpha \rangle. \tag{2.7}$$

Commutation of Θ with the two Virasoro algebras implies that every middle descendant matrix element is this primary matrix element times the two ordinary forms in the last line of (2.6). The analogous statement holds at each outer vertex. Inserting the double-Virasoro decomposition on all four cuts therefore gives the full formula

$$\begin{aligned}
\widehat{\mathbb{F}}_f^{(\eta, \eta')}[\Theta v_{1/2}] &= \sum_{\substack{n_1 \in \frac{1}{2}\mathbb{Z}, n, n', n_3 \in \frac{1}{2}\mathbb{Z} + \frac{1}{4} \\ n' = n \pm \frac{1}{2}, \alpha, \gamma \in \{0, 1\} \\ 2n_1 + \alpha + \gamma \equiv f \pmod{2}}} q_1^{2n_1^2} \hat{q}_{2,1}^{2n^2-1/8} \hat{q}_{2,2}^{2n'^2-1/8} q_3^{2n_3^2-1/8} \eta_1^{2n_1+p_1} \eta_2^\alpha \eta_3^\gamma \\
&\times (-1)^{2n_1+(2n_1+p_1)(\alpha+\gamma)+\alpha\gamma} \frac{\hat{\rho}_f^{(\eta)}(v_{1,n_1}^{p_1}, v_{2,n}^\alpha, v_{3,n_3}^\gamma) \hat{\rho}_f^{(\eta')}(v_{1,n_1}^{p_1}, v_{2,n'}^\alpha, v_{3,n_3}^\gamma)}{\|v_{1,n_1}^{p_1}\|^2 \|v_{2,n}^\alpha\|^2 \|v_{2,n'}^\alpha\|^2 \|v_{3,n_3}^\gamma\|^2} \\
&\times \langle v_{2,n'}^\alpha | \Theta V[v_{1/2}(Q/2)](1) | v_{2,n}^\alpha \rangle \prod_{i=1}^2 F(h_{1,n_1}^{(i)}, h_{2,n}^{(i)}, h_{2,n'}^{(i)}, h_{3,n_3}^{(i)}; h_\psi^{(i)}; c^{(i)} \mid q_1, \hat{q}_{2,1}, \hat{q}_{2,2}, q_3). \tag{2.8}
\end{aligned}$$

The additional split cut supplies one additional squared primary norm. The quadratic part is the original theta sign; $(-1)^{2n_1}$ is the enlarged NS contour sign. The inserted tube operator is even, and both split cuts carry the same total parity. Using squared norms in (2.8) avoids a choice of complex square roots in the numerical contraction.

For comparison with the normalized branching notation, define the radially ordered coefficient, with the NS field in the middle slot, by

$$\mathbb{B}(n', \tfrac{1}{2}, n; \alpha, 1 - \alpha) = \frac{\langle v_{2,n'}^\alpha | V[v_{1/2}(Q/2)](1) | v_{2,n}^{1-\alpha} \rangle}{\|v_{2,n'}^\alpha\| \|v_{1/2}(Q/2)\| \|v_{2,n}^{1-\alpha}\|}. \tag{2.9}$$

No permutation of punctures is implicit in this notation. The two outer normalized coefficients use the original (NS, R, R) order. For $M = 2|n| - \frac{1}{2}$, the raw primary definitions give

$$\Theta v_n^0 = -2^{(-1)^M/2} v_n^1, \quad \Theta v_n^1 = 2^{(-1)^M/2} v_n^0. \tag{2.10}$$

With compatible norm roots such that $\Theta(v_n^\alpha/\|v_n^\alpha\|) = -iv_n^{1-\alpha}/\|v_n^{1-\alpha}\|$, anticommuting Θ through $V[v_{1/2}]$ gives

$$\frac{\langle v_{n'}^\alpha | \Theta V[v_{1/2}](1) | v_n^\alpha \rangle}{\|v_{n'}^\alpha\| \|v_n^\alpha\|} = i \|v_{1/2}\| \mathbb{B}(n', \frac{1}{2}, n; \alpha, 1 - \alpha). \quad (2.11)$$

Accordingly the primary factor in (2.8) is exactly

$$i \|v_{1/2}(Q/2)\| \mathbb{B}_f^{(\eta); p_1}(n_1, n, n_3; \alpha, \gamma) \mathbb{B}_f^{(\eta'); p_1}(n_1, n', n_3; \alpha, \gamma) \mathbb{B}(n', \frac{1}{2}, n; \alpha, 1 - \alpha). \quad (2.12)$$

Here $\|v_{1/2}(Q/2)\|^2 = -Q^2$. The preceding raw formula is the definition when another square-root convention is used.

3 The middle coefficient from the physical Ward identity

The physical stress tensor acts on the SCA factor, whereas (1.4) acts on the auxiliary factor. Hence

$$[L_m, \Theta Q \psi(1)] = 0. \quad (3.1)$$

This statement concerns the physical L_m , not $L_m^{(1)}$ or $L_m^{(2)}$. BPZ conjugation, which is bilinear here, gives

$$\begin{aligned} \langle L_1 v_{n'}^\alpha | \Theta Q \psi(1) | v_n^\alpha \rangle &= \langle v_{n'}^\alpha | \Theta Q \psi(1) | L_{-1} v_n^\alpha \rangle, \\ \langle L_{-1} v_{n'}^\alpha | \Theta Q \psi(1) | v_n^\alpha \rangle &= \langle v_{n'}^\alpha | \Theta Q \psi(1) | L_1 v_n^\alpha \rangle. \end{aligned} \quad (3.2)$$

Use the reusable $\mathbb{V}$ action coefficients from the human notes:

$$\begin{aligned} L_1 v_n^\alpha &= \sum_{|A|+|B|=4n-3} \mathbb{V}_{1,n,\alpha}^{AB} L_{-A}^{(1)} L_{-B}^{(2)} v_{n-1}^\alpha, \\ L_{-1} v_n^\alpha &= \sum_{i=1}^2 \mathbb{V}_{n,\alpha}^{(i)} L_{-1}^{(i)} v_n^\alpha + \sum_{|A|+|B|=4n-1} \mathbb{V}_{-1,n,\alpha}^{AB} L_{-A}^{(1)} L_{-B}^{(2)} v_{n-1}^\alpha \quad (n \geq 3/4). \end{aligned} \quad (3.3)$$

The levels follow from $2n^2 - 2(n-1)^2 = 4n - 2$, followed by the physical level change ∓ 1 . At the boundary $n = 1/4$, $L_1 v_{1/4}^\alpha = 0$ and the second line is replaced by its two same-primary level-one terms and a primary term proportional to $v_{-3/4}^\alpha$. Its coefficient is immaterial below because that term is forbidden by fusion. The action coefficients are inputs from the finite double-Virasoro span decomposition in the human notes; no independent recursion for those coefficients is assumed. When $n' = n + \frac{1}{2} > 0$ and $n \geq \frac{1}{4}$, L_1 on the larger endpoint produces descendants of $v_{n'-1}^\alpha$. In $L_{-1} v_n$ the other-primary branch has label $n-1$ (or $-3/4$ at $n = 1/4$). Its separation from n' is $3/2$, so the middle fusion rule (2.4) kills its contribution. The surviving same-primary descendants obey

$$\rho(\nu_{h'}, \nu_{h_\psi}, L_{-1} \nu_h) = h + h_\psi - h'. \quad (3.4)$$

The first Ward identity thus becomes the scalar recurrence

$$\begin{aligned} & \left[\sum_{i=1}^2 \mathbb{V}_{n,\alpha}^{(i)}(h_n^{(i)} + h_\psi^{(i)} - h_{n'}^{(i)}) \right] \langle v_{n'}^\alpha | \Theta Q \psi(1) | v_n^\alpha \rangle \\ &= \sum_{|A|+|B|=4n'-3} \mathbb{V}_{1,n',\alpha}^{AB} \rho(L_{-A} \nu_{h_{n'-1}^{(1)}}, \nu_{h_\psi^{(1)}}, \nu_{h_n^{(1)}}) \rho(L_{-B} \nu_{h_{n'-1}^{(2)}}, \nu_{h_\psi^{(2)}}, \nu_{h_n^{(2)}}) \langle v_{n'-1}^\alpha | \Theta Q \psi(1) | v_n^\alpha \rangle. \end{aligned} \quad (3.5)$$

If $n = n' + \frac{1}{2} > 0$ and $n' \geq \frac{1}{4}$, apply the second identity instead. Using $\rho(L_{-1} \nu_{h'}, \nu_{h_\psi}, \nu_h) = h' + h_\psi - h$ gives

$$\begin{aligned} & \left[\sum_{i=1}^2 \mathbb{V}_{n',\alpha}^{(i)}(h_{n'}^{(i)} + h_\psi^{(i)} - h_n^{(i)}) \right] \langle v_{n'}^\alpha | \Theta Q \psi(1) | v_n^\alpha \rangle \\ &= \sum_{|A|+|B|=4n-3} \mathbb{V}_{1,n,\alpha}^{AB} \rho(\nu_{h_{n'}^{(1)}}, \nu_{h_\psi^{(1)}}, L_{-A} \nu_{h_{n-1}^{(1)}}) \rho(\nu_{h_{n'}^{(2)}}, \nu_{h_\psi^{(2)}}, L_{-B} \nu_{h_{n-1}^{(2)}}) \langle v_{n'}^\alpha | \Theta Q \psi(1) | v_{n-1}^\alpha \rangle. \end{aligned} \quad (3.6)$$

For negative labels use the human notes' simultaneous reflection $n \mapsto -n$, $P \mapsto -P$ in the action coefficients. Applying L_1 to the endpoint of larger absolute label lowers that absolute label, including the crossing $3/4 \mapsto -1/4$. Every allowed pair reaches one of the two oriented ground pairs. Their values, obtained from (1.7), follow directly from the ground primaries

$$\begin{aligned} v_{\pm 1/4}^0 &= u^0 \otimes w^+ \mp e^{i\pi/4} u^1 \otimes w^-, \\ v_{\pm 1/4}^1 &= \frac{1}{\sqrt{2}} (u^1 \otimes w^+ \pm e^{i\pi/4} u^0 \otimes w^-). \end{aligned} \quad (3.7)$$

With the physical ground metric $\text{diag}(1, i)$ and the auxiliary metric $\text{diag}(1, -1)$, their squared norms are 2 and -1 . Moreover $Q\Theta\psi_0$ maps $v_{\pm 1/4}^0$ to $(Q/\sqrt{2})v_{\mp 1/4}^0$, and $v_{\pm 1/4}^1$ to $-(Q/\sqrt{2})v_{\mp 1/4}^1$. Therefore

$$\begin{aligned} \langle v_{1/4}^0 | \Theta Q \psi(1) | v_{-1/4}^0 \rangle &= \langle v_{-1/4}^0 | \Theta Q \psi(1) | v_{1/4}^0 \rangle = \sqrt{2}Q, \\ \langle v_{1/4}^1 | \Theta Q \psi(1) | v_{-1/4}^1 \rangle &= \langle v_{-1/4}^1 | \Theta Q \psi(1) | v_{1/4}^1 \rangle = Q/\sqrt{2}. \end{aligned} \quad (3.8)$$

No other primary middle matrix elements are needed as anchors. Equations (2.10) and (2.9) then give $\mathbb{B}(n', 1/2, n; \alpha, 1 - \alpha)$.

At a generic momentum the scalar pivot in the recurrence is nonzero. Exceptional zero pivots require continuation of the combined meromorphic expression, not division by zero. The implementation raises an error at such a pivot. Computing the finite ordinary three-point descendant factors in these equations by Ward identities is distinct from computing an entire conformal block. All conformal blocks in (2.8) use the c -recursion described below.

4 The convolution with the physical block

Return temporarily to the tensor-product basis on the split tube. Its physical factor propagates through the identity. A physical BPZ metric at the middle sphere therefore cancels one of the

two physical inverse metrics. Within each physical level and parity this is explicitly $G^{-1}GG^{-1} = G^{-1}$. The remaining physical state has equal level on the two halves and propagation weight

$$\hat{q}_{2,1}^{[B]} \hat{q}_{2,2}^{[B]} = q_2^{[B]}. \quad (4.1)$$

This contracts the physical graph back to the original theta graph. The two contour phases in (2.5) multiply to $(-1)^A$: the NS descendant parity and the total second-slot physical parity agree at the paired vertices, so the two f -dependent signs cancel. This removes the physical part of the enlarged factor $(-1)^{A+A}$. The remaining $(-1)^A$ is precisely the auxiliary NS contour sign in the fermion block. After this linear sign is accounted for, only the quadratic theta signs remain to be compared. The auxiliary factors retain the inserted field and independent levels on the two halves. Because $\Theta\psi$ is even, it identifies the auxiliary parity bits on the two halves. There is consequently no fourth independent parity bit. If ϵ_i and δ_i are the physical and auxiliary parities, respectively, the theta sign at total parity differs from the separate signs by

$$\begin{aligned} \sum_{i < j} (\epsilon_i + \delta_i)(\epsilon_j + \delta_j) - \sum_{i < j} \epsilon_i \epsilon_j - \sum_{i < j} \delta_i \delta_j \\ = \sum_{i < j} (\epsilon_i \delta_j + \delta_i \epsilon_j) \pmod{2}. \end{aligned} \quad (4.2)$$

This gives the original Ramond convolution cocycle:

$$\eta_1^{\epsilon_1} \eta_2^{\epsilon_2} \eta_3^{\epsilon_3} \star_R \eta_1^{\delta_1} \eta_2^{\delta_2} \eta_3^{\delta_3} = (-1)^{\sum_{i < j} (\epsilon_i \delta_j + \delta_i \epsilon_j)} \eta_1^{\epsilon_1 + \delta_1} \eta_2^{\epsilon_2 + \delta_2} \eta_3^{\epsilon_3 + \delta_3}. \quad (4.3)$$

Parity exponents are reduced modulo two. The powers of sewing parameters multiply ordinarily. In particular the complete map on monomials is

$$\begin{aligned} (q_1^a q_2^b q_3^c \eta_1^{\epsilon_1} \eta_2^{\epsilon_2} \eta_3^{\epsilon_3}) \star_R (q_1^{a'} \hat{q}_{2,1}^{b'} \hat{q}_{2,2}^{b''} q_3^{c'} \eta_1^{\delta_1} \eta_2^{\delta_2} \eta_3^{\delta_3}) \\ = (-1)^{\sum_{i < j} (\epsilon_i \delta_j + \delta_i \epsilon_j)} q_1^{a+a'} \hat{q}_{2,1}^{b+b'} \hat{q}_{2,2}^{b+b''} q_3^{c+c'} \eta_1^{\epsilon_1 + \delta_1} \eta_2^{\epsilon_2 + \delta_2} \eta_3^{\epsilon_3 + \delta_3}. \end{aligned} \quad (4.4)$$

Summing this identity over the physical and auxiliary descendants gives

$$\widehat{\mathbb{F}}_f^{(\eta, \eta')}[\Theta v_{1/2}] = \mathbb{F}_f^{(\eta, \eta')}(q_1, \hat{q}_{2,1} \hat{q}_{2,2}, q_3) \star_R \mathbb{F}[\Theta v_{1/2}](q_1, \hat{q}_{2,1}, \hat{q}_{2,2}, q_3). \quad (4.5)$$

This is an identity for the full inserted block. Off-diagonal powers of the two split parameters must be retained on both sides before division. Restricting them to equal powers would replace the inserted field by its zero-mode contribution and would lose terms needed for the convolution.

4.1 The inverse in the required sector

We first establish the sector identity from the physical sewing definition. On either Ramond module use the level-preserving replacement

$$\begin{aligned} \mathbb{L}_{-B} w^+ &\longmapsto (-1)^B e^{-i\pi/4} \mathbb{L}_{-B} w^-, \\ \mathbb{L}_{-B} w^- &\longmapsto -(-1)^B e^{i\pi/4} \mathbb{L}_{-B} w^+. \end{aligned} \quad (4.6)$$

Here B in a sign counts the supercurrent generators, as in the human notes. The map is odd, supercommutes with the physical SCA, and preserves the BPZ pairing with ground metric

$\text{diag}(1, i)$. The last assertion follows at the ground level from $(e^{-i\pi/4})^2 i = 1$ and $(-e^{i\pi/4})^2 = i$; contravariance extends it to descendants. The ground definitions of $\rho_f^{(\eta)}$ give the factor $-i\eta(-1)^{\epsilon_2}$ when this replacement is applied to both Ramond slots. Including the Koszul sign for the two odd maps, the simultaneous action intertwines the Ward identities. The same factor therefore holds for arbitrary descendants, where ϵ_2 is the total physical parity of the second slot.

Applying this change of summation basis at both outer vertices multiplies the unsigned sewing coefficient by $(-i\eta)(-i\eta') = -\eta\eta'$; the two factors $(-1)^{\epsilon_2}$ cancel. Flipping both Ramond parity bits changes the quadratic theta sign by $(-1)^{\epsilon_2+\epsilon_3+1}$. Consequently, at each fixed triple of levels, the coefficient at $(\epsilon_1, 1 - \epsilon_2, 1 - \epsilon_3)$ is $\eta\eta'(-1)^{\epsilon_2+\epsilon_3}$ times the coefficient at $(\epsilon_1, \epsilon_2, \epsilon_3)$. Equation (4.3) also gives

$$(\eta_2\eta_3) \star_R (\eta_1^{\epsilon_1} \eta_2^{\epsilon_2} \eta_3^{\epsilon_3}) = (-1)^{\epsilon_2+\epsilon_3} \eta_1^{\epsilon_1} \eta_2^{1-\epsilon_2} \eta_3^{1-\epsilon_3}. \quad (4.7)$$

Thus $(\eta_2\eta_3) \star_R \mathbb{F}_f^{(\eta, \eta')} = \eta\eta' \mathbb{F}_f^{(\eta, \eta')}$ to every level. In particular, the opposite outer sector satisfies

$$(\eta_2\eta_3) \star_R \mathbb{F}_f^{(\eta, -\eta)} = -\mathbb{F}_f^{(\eta, -\eta)}. \quad (4.8)$$

At zero level, the outer ground forms $\rho_F(\mathbf{1}, u^0, u^0) = 1$ and $\rho_F(\mathbf{1}, u^1, u^1) = i$, together with (1.7), give the auxiliary constant

$$\mathbb{F}_F[\Theta v_{1/2}]|_{q=0} = \frac{Q}{\sqrt{2}}(1 - \eta_2\eta_3). \quad (4.9)$$

This constant has no inverse in the entire parity algebra. However, $(\eta_2\eta_3) \star_R (\eta_2\eta_3) = 1$, so its multiplication on the sector (4.8) is the scalar $\sqrt{2}Q$. For $Q \neq 0$ this scalar is invertible. Thus the original formal inverse is replaced by a sector inverse whose identity is $(1 - \eta_2\eta_3)/2$:

$$\mathbb{F}_F^{-1}[\Theta v_{1/2}] \star_R \mathbb{F}_F[\Theta v_{1/2}] = \frac{1 - \eta_2\eta_3}{2}. \quad (4.10)$$

Within this sector the physical block is uniquely determined by

$$\mathbb{F}_f^{(\eta, -\eta)} = \mathbb{F}_F^{-1}[\Theta v_{1/2}] \star_R \widehat{\mathbb{F}}_f^{(\eta, -\eta)}[\Theta v_{1/2}]. \quad (4.11)$$

The computation need not construct an inverse as a separate series. At each monomial subtract all convolutions involving already determined lower coefficients and positive-level auxiliary coefficients; divide the remainder by $\sqrt{2}Q$. Equation (4.8) makes this triangular division unique. A remainder outside that sector is evidence of incorrect input or insufficient numerical precision and must not be silently projected away.

4.2 Direct fermion sewing: definition and level-ten benchmark

The selected algorithm evaluates the auxiliary factor directly in its fermion basis, while retaining central-charge recursion for both Virasoro factors in the enlarged block. This subsection derives the direct auxiliary construction and records its measured cost for the inserted genus-two graph. The timing below concerns the auxiliary factor alone and performs no comparison with another block; validation of the integrated physical calculation is reported separately.

On the NS edge use the states $\Psi_{-\mathbf{A}}\mathbf{1}$, where $\mathbf{A}$ is a strictly descending list of positive half-integers. On each Ramond edge use $\Psi_{-\mathbf{B}}u^{\mathbf{b}}$, where $\mathbf{B}$ is a strictly descending list of positive integers and $\mathbf{b} = 0, 1$. The symbol $|\mathbf{B}|$ denotes the sum of the modes, whereas $\#\mathbf{B}$ counts them. We retain the conventions

$$\{\psi_r, \psi_s\} = \delta_{r+s,0}, \quad \psi_0 u^{\mathbf{b}} = \frac{u^{1-\mathbf{b}}}{\sqrt{2}}, \quad \psi_r^\dagger = -\psi_{-r}, \quad \langle u^{\mathbf{b}} | u^{\mathbf{b}} \rangle = (-1)^{\mathbf{b}}. \quad (4.12)$$

The Fock metric is diagonal. Its diagonal entries are $(-1)^{\#\mathbf{A}}$ in the NS sector and $(-1)^{\#\mathbf{B}+\mathbf{b}}$ in the Ramond sector.

The incoming state on the additional sphere is at $z_3 = 0$ and carries $\hat{q}_{2,1}^{|\mathbf{B}|}$; the outgoing state is at infinity and carries $\hat{q}_{2,2}^{|\mathbf{B}'|}$. Write their common parity and the parities on the other two edges as

$$\delta_1 = \#\mathbf{A}, \quad \delta_2 = \#\mathbf{B} + \mathbf{b} = \#\mathbf{B}' + \mathbf{b}', \quad \delta_3 = \#\mathbf{C} + \mathbf{c} \pmod{2}. \quad (4.13)$$

Both outer forms vanish unless $\delta_1 + \delta_2 + \delta_3 = 0 \pmod{2}$.

The complete middle insertion. At $z_3 = 1$ every term in $\psi(z_3) = \sum_{m \in \mathbb{Z}} \psi_m z_3^{-m-1/2}$ has unit coordinate factor. Using

$$\Theta \Psi_{-\mathbf{B}} u^{\mathbf{b}} = (-1)^{\#\mathbf{B}+\mathbf{b}+1} \Psi_{-\mathbf{B}} u^{1-\mathbf{b}}, \quad (4.14)$$

the zero-mode action is

$$Q \Theta \psi_0 \Psi_{-\mathbf{B}} u^{\mathbf{b}} = \frac{Q}{\sqrt{2}} (-1)^{\mathbf{b}} \Psi_{-\mathbf{B}} u^{\mathbf{b}}. \quad (4.15)$$

For $k > 0$, creation by ψ_{-k} , when $k \notin \mathbf{B}$, and annihilation by ψ_k , when $k \in \mathbf{B}$, both give the sign $(-1)^{\#\{m \in \mathbf{B}: m > k\}}$ in the descending basis. The resulting word has one more or one fewer fermion. The subsequent Θ action therefore contributes $(-1)^{\delta_2}$ and changes the ground label to $1 - \mathbf{b}$. Consequently the action coefficients of the full insertion are

$$[Q \Theta \psi(1)]_{(\mathbf{B}', \mathbf{b}'), (\mathbf{B}, \mathbf{b})} = \begin{cases} \frac{Q}{\sqrt{2}} (-1)^{\mathbf{b}}, & \mathbf{B}' = \mathbf{B}, \quad \mathbf{b}' = \mathbf{b}, \\ Q (-1)^{\#\{m \in \mathbf{B}: m > k\} + \delta_2}, & \mathbf{B}' = \mathbf{B} \triangle \{k\}, \quad \mathbf{b}' = 1 - \mathbf{b}, \quad k > 0, \\ 0, & \text{otherwise.} \end{cases} \quad (4.16)$$

Here $\triangle \{k\}$ toggles the occupancy of k . For each incoming state, each nonzero mode connects to at most one outgoing Fock state. Both transitions preserve δ_2 , although their two propagation levels need not agree. Reversing a toggle leaves its sign unchanged: the number of occupied modes larger than k and the total parity are unchanged.

The state sum and its signs. In the following formula the subscript on $[Q \Theta \psi(1)]$ means an action coefficient, as in (4.16), rather than a BPZ matrix element. The latter contains the additional outgoing norm $(-1)^{\delta_2}$. The four inverse Fock norms multiply to $(-1)^{\delta_1 + 2\delta_2 + \delta_3}$, so including this outgoing norm leaves $(-1)^{\delta_1 + \delta_2 + \delta_3} = 1$ on every allowed summand. The

NS contour sign and the original theta sign remain. With the two outer forms in the order (NS, R, R), the complete direct definition is therefore

$$\begin{aligned} & \mathbb{F}_F[\Theta v_{1/2}](q_1, \hat{q}_{2,1}, \hat{q}_{2,2}, q_3) \\ &= \sum_{A,B,b,B',b',C,c} q_1^{|A|} \hat{q}_{2,1}^{|B|} \hat{q}_{2,2}^{|B'|} q_3^{|C|} \eta_1^{\delta_1} \eta_2^{\delta_2} \eta_3^{\delta_3} (-1)^{\delta_1 + \delta_1 \delta_2 + \delta_1 \delta_3 + \delta_2 \delta_3} \\ & \times \rho_F(\Psi_{-A} \mathbf{1}, \Psi_{-B} u^b, \Psi_{-C} u^c) \rho_F(\Psi_{-A} \mathbf{1}, \Psi_{-B'} u^{b'}, \Psi_{-C} u^c) [Q\Theta\psi(1)]_{(B',b'),(B,b)}. \end{aligned} \quad (4.17)$$

The ground forms are $\rho_F(\mathbf{1}, u^0, u^0) = 1$ and $\rho_F(\mathbf{1}, u^1, u^1) = i$. They give the constant $Q(1 - \eta_2 \eta_3)/\sqrt{2}$ directly. In particular the second split edge does not introduce an independent parity sign.

The outer forms are computed by moving one fermion contour at a time between the punctures at infinity, one and zero. For the three possible target slots, the useful test sections are

$$z^{k-1/2}(z-1)^{-1/2}, \quad z^{-1/2}(z-1)^{-n-1/2}, \quad z^{-n-1/2}(z-1)^{-1/2}. \quad (4.18)$$

Their expansions give half-integer binomial coefficients and the frame phases $(i, 1, -i)$, together with the usual graded contour signs. For the first section the target NS creation mode is $k - 1/2$; in the other two sections the target Ramond creation mode is n . Remove the largest occupied mode in the first nonempty slot and solve the Ward relation for the original form. Every remaining term has lower total nonzero-mode level, so this procedure terminates at the displayed ground forms. Each local expansion is stopped only after its modes annihilate every state on that leg. The resulting bound is determined by the occupied modes, rather than a fixed expansion order. Computed three-point forms are stored and reused throughout the sum.

Exact rational arithmetic. No irrational arithmetic is needed after extracting $Q/\sqrt{2}$ from the final block. To see this, temporarily use $\sqrt{2}u^1$ in place of u^1 . In this basis the two zero-mode coefficients are $1/2$ and 1 . Strip the known phase i^{δ_2} from the outer form and denote the stored value by ρ_F^{rat} . Thus, with the same fermion words as above,

$$\rho_F = i^{\delta_2} 2^{-(b+c)/2} \rho_F^{\text{rat}}. \quad (4.19)$$

The stored ground values are 1 and 2 . The mode Ward relations then involve only rational binomial coefficients and signs, so every stored value is rational. The two outer phases in a sewn term multiply to $(-1)^{\delta_2}$.

For a diagonal transition, combining these phases, the factor $(-1)^b$ from $\Theta\psi_0$, and the two ground rescalings gives the following normalized summand, apart from the common propagation monomial and sewing sign already displayed in (4.17):

$$\frac{(-1)^{\#B}}{2^{b+c}} (\rho_F^{\text{rat}}(A, B, C))^2. \quad (4.20)$$

For a toggle, $b+b' = 1$. The square root of two in $[Q\Theta\psi(1)]/(Q/\sqrt{2})$ cancels the remaining half-integer ground-rescaling power, while the two factors $(-1)^{\delta_2}$ cancel. Its normalized summand is

$$\frac{(-1)^{\#\{m \in B: m > k\}}}{2^c} \rho_F^{\text{rat}}(A, B, C) \rho_F^{\text{rat}}(A, B', C). \quad (4.21)$$

The ground labels in these abbreviated arguments remain those in (4.17). The temporary basis rescaling has therefore canceled from the physical normalization. The implementation uses exact FLINT rational arithmetic and restores $Q/\sqrt{2}$ only when a numerical value of the full block is requested.

Finite truncation and measured cost. For the balanced split $\hat{q}_{2,1} = u\sqrt{q_2}$, $\hat{q}_{2,2} = u^{-1}\sqrt{q_2}$, total level N means

$$|A| + \frac{|B| + |B'|}{2} + |C| \leq N. \quad (4.22)$$

The stored exponent tuple is $(2|A|, |B|, |B'|, 2|C|)$; its entries sum to at most $2N$. Unequal split levels retain the power $u^{|B|-|B'|}$. At $N = 10$ a single split edge can reach level twenty. To enumerate each nonzero-mode pair once, start from its lower-level state and add the unique toggled mode k . The reverse removal supplies the transposed pair of propagation powers with the same coefficient. The diagonal part is added separately.

A fresh-process run on September 10, 2026, using Python 3.14.3 on macOS arm64, computed the complete normalized auxiliary series through $N = 10$ in 0.186281209s. This construction time includes the provider, Fock bases, Ward cache, all mode transitions, rational normalization and coefficient accumulation. Elapsed process time through imports and the initial result save was 0.290214667s; peak resident memory was 46.765625 MiB. The saved result contains 4793 nonzero exponent tuples and 9586 nonzero rational parity coefficients. The calculation stored 7508 three-point forms and summed 20156 nonzero state contractions. Its coefficients and timing metadata are recorded in `Code/theta_fermion_ccy/results/direct_fermion_level10.json`. The final small rewrite adding timing metadata is excluded from the process-through-save measurement. These measurements establish the cost of this direct auxiliary calculation; they do not constitute a numerical validation or a timing of the complete physical block.

Equation (4.17) gives the full auxiliary factor. Its normalization relative to the reduced central-charge recursion is explained in §5.

Other genera and plumbing channels. The ingredients in (4.17) are local: a fermion Fock basis on each edge, its diagonal pairing, a three-point form on each sphere, and the sparse action of any inserted fermion. They can be sewn on any pants graph with compatible NS and Ramond labels. NS–NS–NS vertices use the corresponding fermion Ward forms; NS–R–R vertices use the forms described above. Spin transports and graded reordering signs are taken from the chosen graph and local coordinates. The particular quadratic theta sign in (4.17) belongs to the present channel and must not be imposed unchanged on another graph. A finite plumbing-level cutoff again leaves a finite sum, and the local three-point data can be reused. This is an arbitrary-genus sewing construction, although the current assembler and the reported runtime concern only the specified four-edge genus-two graph. Its computational cost can grow with the genus and the graph. This direct auxiliary evaluation does not use an Ising Verma-module limit or an Ising central-charge recursion. The two ordinary Virasoro blocks in the enlarged numerator continue to use the prescribed central-charge recursion.

4.3 The Ising decomposition

We keep the auxiliary Ramond parity on each cut. The two irreducible Virasoro modules in the Neveu–Schwarz Fock space have highest weights 0 and 1/2. Each Ramond parity gives an irreducible module of highest weight 1/16. The field inserted on the additional sphere has weight 1/2. The central charge in this subsection is that of the Ising Virasoro algebra, namely $c = 1/2$; it is not the superconformal central charge.

The auxiliary three-point forms in the spin frames of the human notes obey

$$\rho_F(\mathbf{1}, u^0, u^0) = 1, \quad \rho_F(\mathbf{1}, u^1, u^1) = i, \quad (4.23)$$

$$\rho_F(\psi_{-1/2}\mathbf{1}, u^0, u^1) = -\frac{1}{\sqrt{2}}, \quad \rho_F(\psi_{-1/2}\mathbf{1}, u^1, u^0) = \frac{i}{\sqrt{2}}. \quad (4.24)$$

All other ground components vanish by parity. To obtain the second line, move the fermion contour at infinity onto the Ramond puncture at one. The Ward relation gives $i\rho_F(\psi_{-1/2}\mathbf{1}, u^b, u^{1-b}) + \rho_F(\mathbf{1}, \psi_0 u^b, u^{1-b}) = 0$. Substitution of $\psi_0 u^b = u^{1-b}/\sqrt{2}$ gives the displayed values.

The ordered middle insertion acts on a ground state as

$$Q\Theta\psi_0 u^b = \frac{Q}{\sqrt{2}}(-1)^b u^b. \quad (4.25)$$

It commutes with parity and is a primary intertwiner of Ising weight 1/2, since Θ commutes with every auxiliary Virasoro generator. Consequently its descendant matrix elements are precisely those in the Virasoro one-point block. Combining the two outer ground forms with this middle matrix element and the theta sewing signs gives the following sign ledger. The middle row pairing is $Q/\sqrt{2}$ in each row; the signs from its outgoing ground norm have already been included. The norm column is the product of the four internal ground norms, and the last sign includes the NS fermion and theta reordering signs:

h_{NS}	$\mathbf{b}$	$\mathbf{c}$	ρ_F^2	norm product	sewing sign
0	0	0	1	1	1
0	1	1	-1	-1	-1
$\frac{1}{2}$	0	1	$\frac{1}{2}$	1	1
$\frac{1}{2}$	1	0	$-\frac{1}{2}$	-1	1

Thus the complete Ising decomposition is

$$\begin{aligned} \mathbb{F}_F[\Theta v_{1/2}] &= \frac{Q}{\sqrt{2}}(1 - \eta_2 \eta_3) F\left(0, \frac{1}{16}, \frac{1}{16}, \frac{1}{16}; \frac{1}{2}; \frac{1}{2} \mid q_1, \hat{q}_{2,1}, \hat{q}_{2,2}, q_3\right) \\ &\quad + \frac{Q}{2\sqrt{2}}(\eta_1 \eta_2 + \eta_1 \eta_3) q_1^{1/2} F\left(\frac{1}{2}, \frac{1}{16}, \frac{1}{16}, \frac{1}{16}; \frac{1}{2}; \frac{1}{2} \mid q_1, \hat{q}_{2,1}, \hat{q}_{2,2}, q_3\right). \end{aligned} \quad (4.26)$$

Both blocks on the right must use the *irreducible* Ising modules. Their primary coefficients are one. The four internal entries refer, in order, to the Neveu–Schwarz edge, the two parts of the split Ramond edge, and the remaining Ramond edge. The next entry is the external weight and the last is the central charge. The $q_1^{1/2}$ in the second term restores the physical level of the Neveu–Schwarz fermion. This formula contains the complete parity dependence; all remaining coefficients are ordinary scalar Virasoro coefficients.

5 Central-charge recursion and truncation

For generic internal weights, (2.6) is computed with the central-charge recursion of Cho, Collier and Yin [1]. Its poles come from a null state on one of its four internal edges. At a pole on edge e , the internal weight shifts by rs , the corresponding sewing factor contributes q_e^{rs} , and the residue multiplies the lower block by

$$\frac{-\partial_{h_e} c_{rs}(h_e) A_{rs}}{c - c_{rs}(h_e)} P_{rs}[\text{spectators at the first endpoint}] P_{rs}[\text{spectators at the second endpoint}]. \quad (5.1)$$

For the ordered graph (2.6), the two spectator pairs are

e	first pair	second pair
1	(h_3, h_L)	(h_3, h_R)
L	(h_3, h_1)	(h_R, h_ψ)
R	(h_3, h_1)	(h_L, h_ψ)
3	(h_1, h_L)	(h_1, h_R)

The order in each pair matters for odd null level. The external h_ψ stays fixed during the recursion, including when its initial value is degenerate. Replacing it by its degenerate formula evaluated at the recursive central charge would compute a different function.

For completeness the pole and residue data are the ordinary Virasoro ones used in the human notes:

$$\begin{aligned} b_{rs}(h)^2 &= \frac{rs - 1 + 2h \pm \sqrt{(r-s)^2 + 4(rs-1)h + 4h^2}}{1 - r^2}, \\ c_{rs}(h) &= 1 + 6(b_{rs} + b_{rs}^{-1})^2, \\ A_{rs} &= \frac{1}{2} \prod_{\substack{m=1-r, \dots, r, \quad k=1-s, \dots, s \\ (m,k) \neq (0,0), (r,s)}} (mb_{rs} + kb_{rs}^{-1})^{-1}, \\ P_{rs}(h_a, h_b) &= \prod_{\substack{j=1-r, 3-r, \dots, r-1 \\ k=1-s, 3-s, \dots, s-1}} \frac{\lambda_a + \lambda_b + jb_{rs} + kb_{rs}^{-1}}{2} \frac{\lambda_a - \lambda_b + jb_{rs} + kb_{rs}^{-1}}{2}, \end{aligned} \quad (5.2)$$

where $h_a = ((b_{rs} + b_{rs}^{-1})^2 - \lambda_a^2)/4$ and likewise for h_b . The sum runs over $r \geq 2, s \geq 1$. The numerical continuation chooses the noncancelling quadratic root (the larger absolute value), consistently across (r, s) and (s, r) . This matters at negative weights when $s = 1$, where a cancelling root can be zero. Unresolved confluent poles raise an error; a singular residue is never simply discarded.

The regular term at large c is the global $SL(2)$ block times the universal large- c vacuum factor. The global coefficient at levels (a, b, c, d) is

$$\frac{\rho(L_{-1}^a \nu_{h_1}, L_{-1}^b \nu_{h_L}, L_{-1}^d \nu_{h_3}) \rho(L_{-1}^a \nu_{h_1}, L_{-1}^c \nu_{h_R}, L_{-1}^d \nu_{h_3}) \rho(L_{-1}^c \nu_{h_R}, \nu_{h_\psi}, L_{-1}^b \nu_{h_L})}{a!(2h_1)_a b!(2h_L)_b c!(2h_R)_c d!(2h_3)_d}. \quad (5.3)$$

The universal factor is independent of all light weights. We may therefore forget the external puncture and collapse the extra sphere. Every sewing and collapsing map is Möbius, so

no Schwarzian term is introduced. Equation (1.2) shows that the factor depends only on $(q_1, \hat{q}_{2,1}\hat{q}_{2,2}, q_3)$. It is computed by the leading Gaussian contractions of $L_{-m}/\sqrt{c}$, $m \geq 2$. For multiplicities k_m the diagonal norm is

$$\left\| \prod_{m \geq 2} (L_{-m}/\sqrt{c})^{k_m} \mathbf{1} \right\|^2 \longrightarrow \prod_{m \geq 2} k_m! \left[\frac{m(m^2 - 1)}{12} \right]^{k_m}. \quad (5.4)$$

The pair contractions between the ordered slots are

$$\begin{aligned} C_{\infty,0}(m, n) &= \delta_{mn} \frac{m(m^2 - 1)}{12}, \\ C_{\infty,1}(m, n) &= \frac{m(m^2 - 1)}{12} \binom{m-2}{n-2} \quad (m \geq n), \\ C_{1,0}(m, n) &= \frac{(-1)^m (m+n-1)!}{12(m-2)!(n-2)!}. \end{aligned} \quad (5.5)$$

The second is zero when $m < n$, and same-slot contractions are zero. These formulas follow by expanding or differentiating $\langle T(z)T(w) \rangle = c/[2(z-w)^4]$ at the three punctures. At a fixed triple of vacuum levels, sum the square of the three-slot Wick contraction over oscillator partitions and divide by the three diagonal norms above. The code performs this finite sum exactly over the rationals. It uses the large- c algebra, not a finite- c Gram inverse. The reduced double-Virasoro numerator is missing two copies of this factor. The selected direct fermion sum contains its full sewing dependence, so division by it must be followed by restoring two copies in the physical result.

To define a finite physical total-level truncation without placing the insertion at an endpoint, write

$$\hat{q}_{2,1} = u\sqrt{q_2}, \quad \hat{q}_{2,2} = u^{-1}\sqrt{q_2}. \quad (5.6)$$

A term $q_1^{a/2} \hat{q}_{2,1}^l \hat{q}_{2,2}^r q_3^{d/2}$ has physical total degree $(a+l+r+d)/2$ and insertion-position dependence u^{l-r} . Through physical total level N retain all terms with $a + l + r + d \leq 2N$. This is a downward-closed truncation, so the triangular convolution uses no coefficients above the cutoff. The physical answer must have $l = r$ after division, but this property is a result to be checked, not a restriction imposed on the computation. At $N = 10$ the contributing middle primary pairs satisfy $|n|, |n'| \leq 9/4$. For the middle recurrence the reusable L_1 and relevant L_{-1} coefficients require descendant levels at most six. This bound does not apply to the entire outer Ward grid: the current outer implementation also constructs $L_{-1}v_{9/4}$, whose other-primary branch reaches descendant level eight. For negative branch labels, the implementation first uses the reflection $v_{-n}(P) \leftrightarrow v_n(-P)$ and reverses the labels of every term in the action expansion. This is an intertwiner of the physical SCA and both Virasoro algebras. On the physical Ramond grounds it acts as $\text{diag}(1, -1)$ and maps the defining fermion strings without an extra normalization factor. Consequently the action coefficients can be solved on the positive-label chart; no high-level Fock-to-PBW reflection matrix is needed. The action data are stored for each momentum and reused across the outer Ward equations.

6 Implementation and numerical status

The implementation is in `Code/theta_fermion_ccy`. `middle_branching.py` implements (3.5)–(3.6); `outer_branching.py` uses the existing outer L_1 Ward system; `punctured_ccy.py` supplies genuine central-charge residues and the global seed; `direct_fermion.py` evaluates the full auxiliary factor by its mode Ward identities and diagonal Fock pairing; `pipeline.py` performs the branch sum and `series_algebra.py` performs the sector division. The archived outer oracle uses the opposite label for an odd form: $\eta_{\text{native}} = (-1)^f \eta_{\text{physical}}$. The adapter performs this transport before solving its Ward system. It follows algebraically from comparing the two $G_{-1/2}$ Ward conventions; the numerical benchmark below tests $f = 0$ only.

Only the two requested numerical comparisons are designated as validation: the final physical block against an independent physical PBW calculation through total level five, and the final physical block at different split parameters with their product held fixed. The PBW reference is stored separately and is not imported by the production pipeline. No additional component numerical comparison suite is used.

The two requested comparisons pass through level five. The benchmark is $b = 7/5$, $(P_1, P_2, P_3) = (11/23, 13/29, 17/31)$, $p_1 = f = 0$ and $(\eta, \eta') = (+, -)$. The calculation uses 70 decimal digits for the outer Ward system, 100 for the CCY engines, and exact rational fermion sewing before restoring its common $Q/\sqrt{2}$ factor. The independent physical PBW reference uses 384-bit FLINT arithmetic. All 4,648 parity-coefficient slots in the four-variable truncation are compared, including off-diagonal slots whose physical answer is zero. The maximum absolute difference is 4.72×10^{-52} ; the maximum difference divided by $\max(1, |\text{PBW coefficient}|)$ is 4.72×10^{-52} .

The split-position comparison uses $(q_1, q_2, q_3) = (0.013, 0.007, 0.009)$, $u = 1/2, 1, 2$, and all eight evaluations of the spin-lift signs. The maximum change, divided by $\max(1, |\mathbb{F}(u = 1)|)$, is 2.96×10^{-60} . For example, at $(\eta_1, \eta_2, \eta_3) = (1, -1, 1)$ all three values agree with 2.0178804284272157747114188332429722451467 to the digits shown. These are numerical differences, not interval error bounds.

The pipeline timer includes all mathematical stages and intermediate checkpoint writes. It starts after imports and argument parsing and ends before encoding and saving the final physical series. This level-five run took 82.1968 seconds in total: 52.1231 for the outer coefficients, 2.6851 for the middle recurrence, 26.1996 for the two Virasoro blocks in the branch sum, 0.0825 for the direct auxiliary factor, and 0.6678 for division and the restored vacuum factors. The balance is assembly and intermediate serialization. This is a *level-five* runtime; it is not a level-ten estimate. The complete output and the two comparisons are saved under `Code/theta_fermion_ccy/results` as `ccy_direct_reflected_level5.json` and `validation_ccy_direct_reflected_level5.json`.

The production calculation retains the input, anchor and Ward-solution precision throughout. Its sector guard raises an error if division leaves the required ideal; it does not project the answer to force the identity. The separate level-ten auxiliary benchmark in §4.2 is not a runtime for the physical block. The full level-ten production timing is being measured with the same implementation and precision settings.

References

- [1] M. Cho, S. Collier and X. Yin, *Recursive Representations of Arbitrary Virasoro Conformal Blocks*, arXiv:1703.09805.
- [2] N. Javerzat, R. Santachiara and O. Foda, *Notes on the solutions of Zamolodchikov-type recursion relations in Virasoro minimal models*, arXiv:1806.02790.